# ERGONOMICS IN HCI

Introduction • Real-World Examples • Case Study • Class Activity



# Learning Objectives

## By the end of this lecture, students can...

- Define ergonomics in simple words.
- Explain the connection between ergonomics and HCI.
- Identify physical, cognitive, and environmental factors.
- Recognize ergonomic problems in everyday technology.
- Suggest practical improvements using a case study.



**Key idea**

Design technology around human needs — not the other way around.

# What is Ergonomics?

The science of fitting work, tools, systems, and environments to people.



## HUMAN

Body, mind, abilities, limitations



## TECHNOLOGY

Computer, keyboard, screen, software, mobile apps



## ENVIRONMENT

Lighting, noise, temperature, workspace

**ERGONOMICS = FIT TECHNOLOGY TO PEOPLE**

# What is HCI?

## Human–Computer Interaction




Examples: typing, clicking, touchscreens, websites, ATMs, mobile apps.

# Ergonomics + HCI

## The central idea of this chapter




**Is the interaction comfortable, safe, easy and effective?**

# Why Does Ergonomics Matter?

### Comfort

Users should not fight the technology.

### Safety

Reduce avoidable physical strain.

### Efficiency

Help users complete tasks smoothly.

### Satisfaction

A comfortable interaction supports better UX.

**Poor design → discomfort → fatigue → errors → poor experience**

# Three Major Areas of Ergonomics



## 1. Physical Ergonomics

Body posture • movement • chair • keyboard • monitor • reach



## 2. Cognitive Ergonomics

Memory • attention • perception • mental workload • decisions

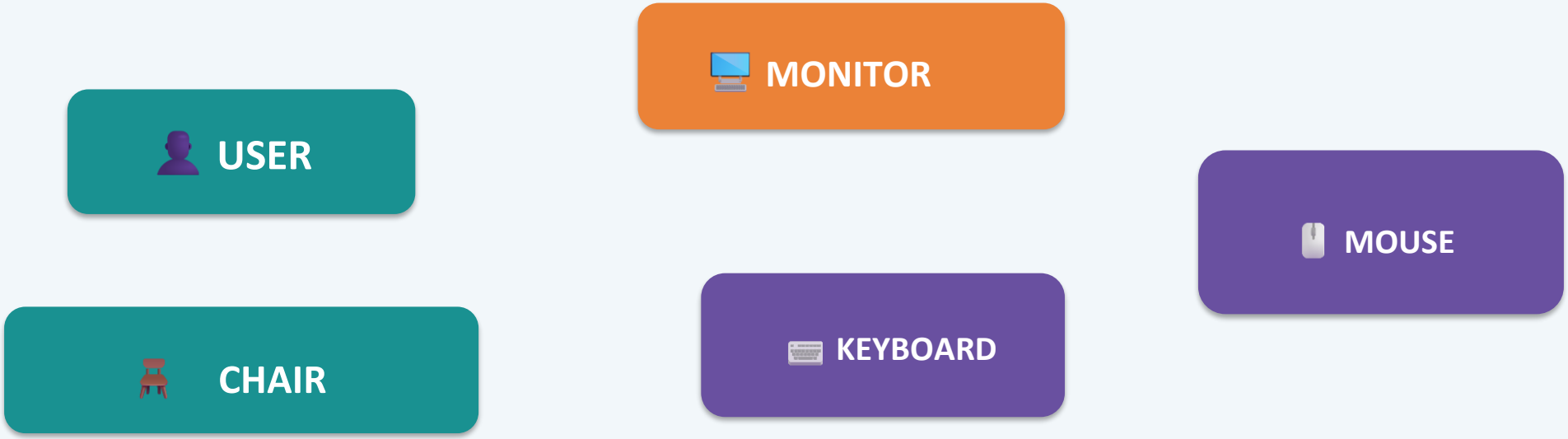


## 3. Environmental Ergonomics

Lighting • glare • noise • temperature • workspace

# Real-World Example #1: Computer Workstation

## Look for the ergonomic problems




Ask: Is the screen too low?

Is the chair supportive?

Are keyboard and mouse within easy reach?

Is lighting causing glare?

# Real-World Example #2: ATM

Ergonomics is not only about chairs!

↔

## Reach

Buttons should be reachable.



## Visibility

Text and instructions should be readable.

✓

## Feedback

The ATM should clearly show what happened.

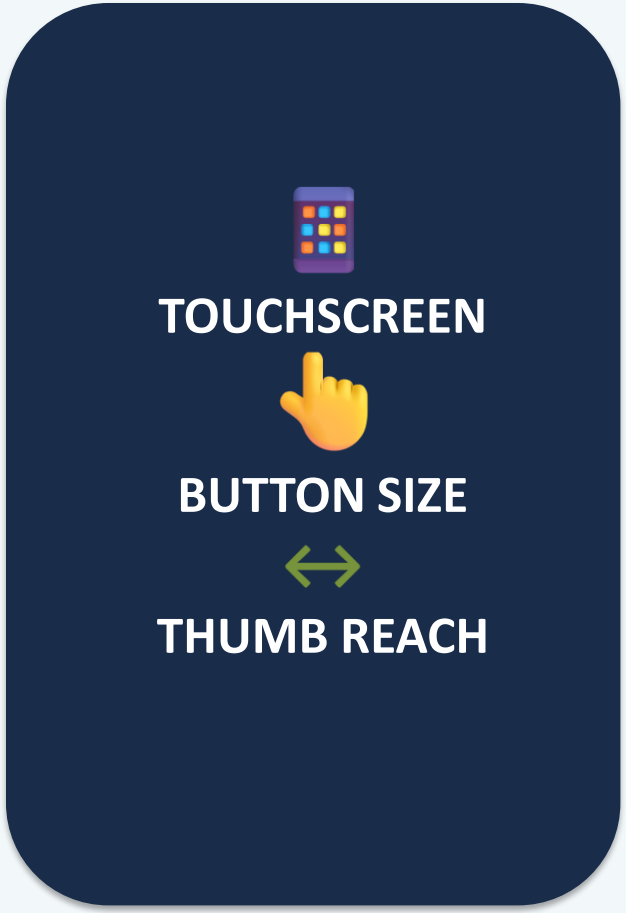


## Privacy

The interaction should protect sensitive actions.

# Real-World Example #3: Smartphone

## Mobile HCI must consider hands, fingers, reach, and attention.




Large enough touch targets reduce accidental taps.

Important controls should be easy to reach.

Clear spacing improves accuracy.

Simple layouts reduce mental effort.

# Real-World Example #4: Car Dashboard

A driver cannot spend long periods searching for controls.



### 👀 Visibility

Important information should be easy to notice.



### ✋ Reach

Frequently used controls should be accessible.



### 🧠 Cognitive Load

Avoid unnecessary complexity while driving.

# Good vs Poor Ergonomics

| POOR ERGONOMICS | BETTER ERGONOMICS |
| --- | --- |
| Monitor too low | Monitor positioned appropriately |
| Tiny / crowded controls | Clear and appropriately sized controls |
| Uncomfortable chair | Supportive adjustable seating |
| Strong glare / poor lighting | Comfortable, controlled lighting |

# Case Study: University Computer Lab

## Scenario for group discussion



### THE PROBLEM

Students use computers for several hours. They report neck discomfort, eye strain, wrist discomfort, and difficulty concentrating.



### OBSERVATIONS

- Monitors are too low
- Chairs are uncomfortable
- Sunlight causes screen glare
- Keyboards are far away
- Lab software has small, crowded buttons

# Case Study: Your Task

## Work in groups of 3–4

**1**

**IDENTIFY**

Find 4 ergonomic problems

**2**

**CLASSIFY**

Physical, cognitive, or environmental?

**3**

**EXPLAIN**

Why does each problem matter?

**4**

**REDESIGN**

Give one practical solution for each

**Time: 8 minutes • Prepare a 2-minute group response**

# Class Activity: Ergonomics Detective 🔎

Observe your own classroom or computer lab



**LOOK**

Find 3 ergonomic problems.



**RECORD**

Write what you observed.



**SOLVE**

Suggest a practical improvement.

**Challenge: Can your group find a problem that others missed?**

# Quick Check

## Discuss with a partner

1. Why is ergonomics important in HCI?
2. Give one physical ergonomic example.
3. Give one cognitive ergonomic example.
4. Give one environmental ergonomic example.
5. Name one ergonomic problem in a smartphone app.

# Key Takeaways

**ERGONOMICS = FIT PEOPLE + TECHNOLOGY**



## Remember

Comfort • Safety • Efficiency • Satisfaction



## Look Beyond Hardware

Ergonomics also affects software, mobile apps, touchscreens, and environments.



## Think Like a Designer

Ask: What does the human need, and what can we change?

NEXT SUBTOPIC

# Human Factors in Ergonomics

Physical capabilities & limitations

Vision, hearing and movement

Attention, perception and memory

Individual differences